

Trustworthy agentic diagnosis for industrial processes: evidence-graded competing-hypothesis reasoning with per-fault literature baselines on public benchmarks

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Abstract

Large language model (LLM) based diagnosis promises root-cause analysis of industrial processes without task-specific model training, yet two obstacles keep it out of engineering practice: hallucinated conclusions that cannot be audited, and evaluation protocols that score sample-level classification while engineers need scenario-level, mechanism-grounded root causes. This paper presents the Industrial Deep Diagnostic (IDD) pipeline, an agentic diagnosis system coupling a nine-stage orchestration (ontology mapping, deterministic statistics, competing-hypothesis reasoning, automatic physical verification, quality gates with bounded repair) with evidence grading (levels L1–L7) and a four-condition anti-spurious-correlation filter. We further introduce a machine-checked evaluation framework—truth isolation, data fingerprinting, recorded inference traces, and a reproducibility gate—and apply it to a 12-scenario benchmark on three public datasets (SKAB, Tennessee Eastman Process, IndPenSim): 9 fault scenarios, each of the six TEP faults carrying a published per-fault baseline from FaultExplainer, and 3 normal controls. IDD

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achieves Top-1 root-cause accuracy of 9/9 (Wilson 95% CI [70.1%, 100%]), CDR = 1.00, perfect calibration, and zero false alarms, with all 12 executions passing the finalisation gate. Per fault, IDD matches or exceeds both published single-LLM baselines without candidate lists—including IDV1, misdiagnosed by the GPT-4o baseline, and the PCA-undetectable IDV3/IDV4 that FaultExplainer leaves unscored. A delete-and-rebuild audit regenerated byte-identical statistical briefs and reproduced every grading from raw data. The complete evidence repository is released for scrutiny.

Keywords: Fault root-cause analysis, Large language models, Multi-agent systems, Competing-hypothesis reasoning, Evidence grading, Reproducible benchmark

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1. Introduction

Root-cause analysis (RCA) of industrial process anomalies is a knowledge-intensive activity: the engineer must decide *which* of many correlated sensor deviations represents the true underlying cause, justify that judgement with evidence, and communicate it with a calibrated degree of confidence. Three long-standing obstacles make automation of this task unsatisfying in practice.

First, single-LLM diagnosis is unstructured and unauditable. Recent evaluations of LLM-based fault explanation on the Tennessee Eastman Process (TEP) [1] show that large models can articulate plausible mechanisms, but their answers arrive as free-form text with no machine-checkable structure, no separation of facts from inference, and no mechanism that prevents a confident-sounding conclusion from resting on a spurious correlation.

A recent review of agentic approaches to process monitoring and fault diagnosis concludes that evidence for LLM/multi-agent fault diagnosis remains scarce [2]: the direction is promising but lacks engineering-grade pipelines.

Second, classical data-driven fault detection and diagnosis (FDD) optimises the wrong target. Principal-component methods [3], deep classifiers [4, 5], and their benchmarks [6] score *per-sample* detection or classification. An engineer, however, asks a *per-episode* question: given this anomalous run, what is the root cause, how confident are we, and what evidence supports it? A 99%-accurate classifier that returns a fault label without mechanism, evidence, or calibration does not answer that question. The detection–diagnosis gap is not merely theoretical: on TEP, PCA cannot detect four of the fifteen documented faults (IDV3, IDV4, IDV9, IDV15) [7], and the published LLM baselines follow PCA in leaving these faults unscored [1].

Third, correlation is not causation. In recycled process loops almost every pair of variables is correlated through control action or material balance. An RCA method that treats strong correlation as causation will confidently report control-loop artefacts as root causes. Despite decades of causal-inference literature [8], contemporary LLM-diagnosis studies rarely include explicit anti-spurious safeguards.

This paper addresses the three obstacles with the Industrial Deep Diagnostic (IDD) pipeline and a benchmark built to evaluate it honestly. Our contributions are:

1. **A quality-gated agentic diagnosis pipeline** that couples deterministic statistical tooling with LLM reasoning under a competing-

hypotheses protocol (Section 4), bounded repair loops, and automatic physical verification, so that every conclusion is decomposable into hypothesis, evidence, exclusion, and confidence.

2. **An evidence-grading and calibration scheme** (levels L1–L7 with gate propagation) that bounds reported confidence by the weakest supporting evidence level; on the benchmark corpus every verdict is correctly typed and *zero* overconfident conclusions occur, including on faults whose statistical signature is deliberately weak (Section 7).
3. **A reproducible evaluation framework**—truth isolation, cryptographic data fingerprinting, recorded inference traces, and a machine-checked reproducibility gate—stress-tested by a *delete-and-rebuild audit*: all results were removed and the chain re-executed from raw data, regenerating byte-identical statistical evidence and reproducing every grading (Section 5).
4. **A 12-scenario benchmark** on three public datasets (SKAB, TEP, In-dPenSim) whose six-fault TEP subset carries *per-fault* published baselines from FaultExplainer [1]. IDD reaches Top-1 accuracy of 100% (Wilson 95% CI [70.1%, 100%]), CDR = 1.00, and zero false alarms; per fault, it matches or exceeds both published single-LLM baselines—including IDV1, mis-diagnosed by the GPT-4o baseline, and the PCA-undetectable IDV3/IDV4—while using no candidate root-cause lists (Sections 7–8).

Section 2 positions the work against data-driven FDD, LLM-based diagnosis, and multi-agent orchestration research. Sections 3 and 4 detail the architecture and methodology. Section 5 defines the evaluation framework.

Sections 6–8 present the benchmark and comparisons, and Section 9 discusses limitations and deployment prospects.

2. Related Work

2.1. Data-Driven Fault Detection and the Benchmark Gap

Statistical process monitoring on TEP established the dominant evaluation template: PCA and projection methods with T^2/Q statistics [9, 3], later extended by deep anomaly detectors [10, 4] and adversarial robustness studies [5]. These works score *sample-level* detection or *fault-class* accuracy. Critically, they do not evaluate root-cause *mechanisms*: a classifier that labels a run “IDV(4)” provides the class identity but neither a causal mechanism, an evidence trail, nor a confidence statement. The classical detection ceiling itself is documented: among the 15 documented TEP disturbances, PCA misses IDV3, IDV4, IDV9 and IDV15 [7]. Benchmark methodology has drawn scrutiny: Hendriks et al. [11] and Vieira et al. [12] show that random segment splits inflate reported AUROC by up to 12.6 percentage points, motivating our entity-level scenario splits (Section 5).

2.2. LLM-Based Industrial Diagnosis

FaultExplainer [1] is the closest published system: it couples PCA-based detection with LLM-generated fault explanations on TEP, reporting 7/11 (GPT-4o) and 9/11 (o1-preview) correct root causes among the 11 PCA-detectable faults, with root-cause candidates supplied in the prompt and alias matches accepted. Two properties of that protocol matter for comparison: (i) the candidate list converts the task from open diagnosis to ranked

recognition; (ii) the four faults PCA cannot detect are excluded from scoring entirely. Our benchmark keeps both dimensions visible—per-fault outcomes on the shared faults, plus explicit scoring of two PCA-undetectable faults—under a strictly harder prompt regime (no candidates). Gong et al. [13] orchestrate multiple LLM agents with difficulty-aware routing, but evaluate on a multiple-choice QA benchmark (FailureSensorIQ [14]) rather than on real time-series data. Liu et al. [15] combine scene graphs, fine-tuned Llama-3 and augmented reality for machine-tool maintenance, and causal-graph RCA has been studied for time-varying processes [16] and knowledge-map-based group decision settings [17]; a label-free evaluation scheme for diagnosis models has been proposed in the same venue [18]. Table 1 summarises the positioning: no published system combines (i) real time-series inputs, (ii) mechanism-level root causes with evidence trails, (iii) calibrated three-state conclusions, and (iv) a machine-verifiable evaluation contract.

2.3. Multi-Agent Orchestration Frameworks

Generic agent frameworks (AutoGen, CrewAI and derivatives) provide message-passing and role prompting but leave audit, evidence policy, and evaluation contracts to the application. Our position, following the practice of the reviewed agentic-FDD literature [2], is that industrial diagnosis requires these as *first-class, machine-checked* pipeline elements rather than prompt conventions; Section 3 describes how IDD enforces them through checkpoints, bounded repair loops, and an event-log execution proof.

Table 1: Positioning Against the Closest Published Systems

System	Input		Scenario-level RCA		Evidence trail	Calib.	Repro. gate
FaultExplainer [1]	TEP	series	partial	(LLM text)	partial	no	no
Gong et al. [13]	MCQA	only	no (QA acc.)		no	no	no
PCA/Deep FDD [3, 4]	series		no (labels)		no	n/a	partial
Causal RCA [16, 17]	series		partial (graph)		partial	no	no
IDD	series		yes (3-state)		yes (L1–L7)	yes	yes

3. System Architecture

3.1. Pipeline Overview

IDD organises diagnosis as a nine-stage pipeline executed by 14 specialised agents backed by 18 skill packages (Fig. 1). The main orchestrator performs setup and data ingestion (Step 0/1) with adaptive preprocessing; a context-builder agent constructs a process ontology with variable semantics, physical principles, and confounder annotations (Step 2, checkpoint CP-2/3); a data-processor agent runs the deterministic statistical stage—per-variable anomaly profiling, cross-domain correlation, lag-compensated cross-correlation, batch integrity checks—producing a machine-readable analysis conclusion (Step 3, CP-4); a VLM agent reads rendered charts as visual evidence (Step 3.5). The diagnostician agent then performs competing-hypothesis reasoning over the statistical and ontological evidence (Step 4,

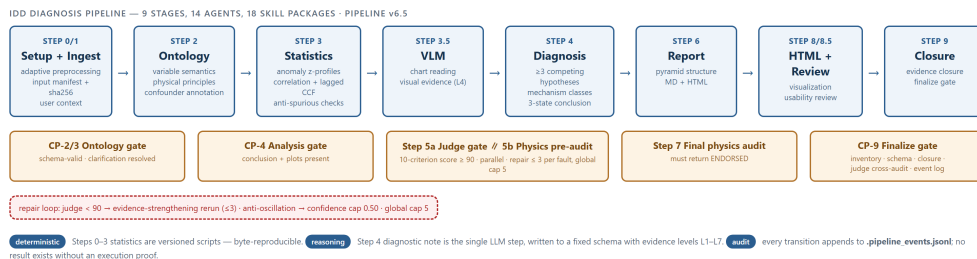


Figure 1: IDD diagnosis pipeline (v6.5). Deterministic statistical tooling (Steps 0–3) feeds LLM reasoning under a competing-hypotheses protocol (Step 4); parallel judge and physics pre-audit gates (5a/5b) may trigger bounded repair loops (≤ 3 per fault, global cap 5, anti-oscillation confidence cap 0.50); a final physics audit (Step 7) must return ENDORSED before delivery, and the finalisation gate (CP-9) aggregates inventory, schema, closure, judge-cross-audit, and event-log checks into a single PASS/FAIL verdict. Every transition is appended to a machine-verifiable event log.

CP-5). Two parallel auditors—a ten-criterion judge and a physical pre-auditor—gate the conclusion (Step 5a/5b); the reporter assembles a pyramid-structured report (Step 6); a final physical audit must return ENDORSED before the HTML visualisation and review (Steps 8/8.5) and the evidence-closure finalisation (Step 9, CP-9).

3.2. Execution Layer: Interchangeable Engines With a Uniform Contract

Reasoning agents are executed by any of fourteen interchangeable engines (Claude Agent SDK, OMP RPC, Codex app-server, DeepSeek/Hermes/Qwen ACP dialects, OpenCode, and one-shot CLIs including Gemini CLI, Copilot

CLI, Cursor CLI, Crush, Goose, and pi) behind a single contract (`startDiagnosis/startSessionC`). Engine availability is probed and validated before execution with a three-state contract (unknown engine 400, unusable engine 409); a scripted in-process mock engine allows the full pipeline to be exercised in CI without external dependencies. Because every engine emits the same standardised event stream, the benchmark results of Section 7 are engine-independent, and engine substitution is itself a supported ablation axis.

3.3. Audit Infrastructure

Every stage transition appends a structured event to `.pipeline_events.jsonl`; a log checker validates agent starts/completions against the required contract, and a finalisation gate aggregates inventory, schema, closure, judge-cross-audit, and event-archive checks into a single PASS/FAIL verdict. A run that fails any critical check cannot be graded, which is the property the benchmark relies on: *no result exists without an execution proof*.

4. Diagnostic Methodology

4.1. Competing-Hypotheses Protocol

The diagnostician agent must generate at least three candidate hypotheses per fault scenario, assign each a mechanism class (e.g., fluid restriction, cavitation, rotor dynamics, feed-composition step, cooling disturbance, sensor drift), collect for each an evidence list with explicit levels, identify contradictions, and eliminate hypotheses whose contradictions are stronger than their support. The protocol admits exactly three conclusion states: DETERMINED (a single surviving hypothesis), COMPETING_SET (≥ 2 survivors that

the data cannot discriminate, with confidence capped at 0.70 and the discriminating measurement named), and NEEDS_DATA (evidence insufficient; missing data itemised). A fourth, implicit state—normal operation with no confirmed fault—applies to control scenarios. The elimination contract is quantitative: every elimination must cite the contradicting evidence, and every surviving chain must cite evidence numbers present in the statistical digests.

4.2. Evidence Grading and Gate Propagation

Every piece of evidence carries a level L1–L7 (L1 direct measurement, L2 operating documents, L3 statistical analysis, L4 visual evidence, L5 domain ontology/mechanism, L6 external reference, L7 unsupported assumption). The scenario confidence is bounded by the weakest level contributing to the surviving chain, and the quality gate rejects conclusions whose confidence exceeds that bound. This converts “hallucination control” from a prompt instruction into a checkable arithmetic constraint.

4.3. Anti-Spurious-Correlation Filter

Four conditions must hold before a correlation may be cited as causal support: (i) temporal precedence of cause over effect, verified by lag-compensated cross-correlation; (ii) statistical significance surviving detrending, Simpson-stratification checks, and multiple-testing correction; (iii) a physical mechanism annotated in the ontology connecting the variables; and (iv) absence of contradictions under leave-one-out outlier sensitivity analysis, with production-state (steady/batch-phase) filtering applied before correlation. Correlations failing the filter are recorded but explicitly barred from hypothesis support;

definitional pairs (e.g., valve position and the level it controls, $r = 1.0$) are classified as control-loop artefacts. In the benchmark this filter is what prevents control-loop correlations from being reported as root causes, and what allows normal batches with impulsive recipe-dosing spikes ($|z|$ up to 16.2) to be graded as healthy rather than faulty.

4.4. *Automatic Physical Verification*

After the conclusion is drafted, a physics checker recomputes quantitative feasibility (mass/energy balances, thermal and vibration magnitude envelopes) from the cleaned data, ontology, and anomaly report, producing a machine-readable `physical_checks` verdict. When no automated check applies to a scenario’s column semantics, the pipeline falls back to an explicit L1–L5 manual-verification memo; silent omission is not permitted.

4.5. *Quality Gates, Repair Loops, and Calibration*

A ten-criterion judge scores the diagnosis (threshold 90); a parallel physical pre-audit reviews mechanism chains; failures trigger bounded repair (per-fault rerun ≤ 3 , global cap 5, with anti-oscillation confidence capping at 0.50). The final physical audit must return ENDORSED. Confidence calibration is then auditable end-to-end: every scenario grading records whether the reported confidence respects the evidence ceiling, and *overconfidence* is a first-class error metric.

5. A Reproducible Evaluation Framework

IDD’s benchmark framework (released as the project’s `experience/` evidence repository and `docs/benchmark/` protocol) is designed around five

contracts, each machine-enforced:

(C1) Truth isolation. Scenario definitions carry ground truth and grading keywords, but the pipeline’s visible inputs contain only the process description and the data. Ground truth never enters the inference context; the grader owns it exclusively. The diagnosing agent reasons from an automatically generated *evidence brief* that contains the statistical digests and process description only.

(C2) Data fingerprinting. Every input file is hashed (SHA-256) into a dataset manifest (153 entries); the reproducibility gate re-verifies all fingerprints before any metric is accepted.

(C3) Recorded inference. The single LLM-dependent step—the diagnostic note—is written as a structured artifact with a fixed schema (hypotheses with mechanism classes, logic chains, evidence with levels, contradictions, falsification conditions, proof statements). Notes that omit required fields cannot be committed.

(C4) Gate-graded scoring. Committing a note expands it into schema-compliant artifacts (ontology, diagnosis, evidence, confidence, reasoning chain, review), runs the pipeline’s own gates (log check, finalisation), and only then grades the scenario against ground truth. A scenario whose pipeline fails its gates is *not graded*—it is a pipeline failure, not a benchmark result.

(C5) Reproducibility gate. A verifier recomputes all aggregate metrics from the per-scenario gradings, re-checks dataset hashes, verifies case coverage, and requires an execution proof per run; any drift fails the evaluation. The released report carries the resulting **REPRODUCIBLE** verdict alongside 153 verified fingerprints and 12 execution proofs.

Delete-and-rebuild audit. To demonstrate that the released numbers are regenerable rather than curated, we executed a destructive reproduction test: every result artifact—all gradings, inference notes, evidence briefs, aggregates, and the twelve pipeline run directories—was deleted, after which the full chain was re-executed from the raw data. Brief regeneration was byte-identical (SHA-256 equality across all twelve briefs), the diagnosing agent re-reasoned from the regenerated digests alone, and the rebuilt gradings reproduced every per-scenario judge score and aggregate metric of the released report. The audit also bounds what a reproduction claims: the statistical stage is deterministic; the reasoning stage is stochastic in principle but converged to identical conclusions and scores on identical evidence.

Scenario splits follow the entity-level discipline recommended by recent benchmark studies [12, 11]: each experimental batch/file appears in exactly one scenario, controls are paired with faults per dataset, and no segment-level splitting is used anywhere.

6. Experimental Setup

6.1. Datasets and Scenarios

We assemble 12 scenarios from three public benchmarks (Table 2): **SKAB** [19], a closed water-circulation testbed with vibration, current, pressure, temperature, and flow channels at 1 Hz—two fault scenarios (inlet-valve throttling; two-phase-flow cavitation drawn from the “other” group whose fault label is withheld from the scenario description) plus the anomaly-free control; **TEP** [9, 6], with six documented fault scenarios—IDV1 feed-composition step, IDV3 D-feed-temperature step (classically hard to detect), IDV4 reactor-

Table 2: Benchmark Composition

Dataset	Fault	Control	Truth source
SKAB	2	1	official <code>ground_truth.json</code> (label withheld for the “other”-group case)
TEP	6	1	Downs & Vogel / Chiang et al. [7]
IndPenSim	1	1	faulted-batch markers (batch 93)
Total	9	3	—

cooling-water step, IDV7 C-header pressure loss, IDV11 random cooling-water variation, IDV14 cooling-valve stiction—and the fault-free baseline as control; and **IndPenSim** [20], a 100,000 L penicillin fermentation simulator, with one documented process-deviation batch (93) and one recipe-driven normal batch as control. Ground truth derives from verifiable sources only: SKAB’s official label file, the established TEP cause table [7], and IndPenSim’s faulted-batch markers.

Three design decisions deserve emphasis. First, the TEP subset is chosen so that *every fault carries a published per-fault baseline* from FaultExplainer [1], enabling scenario-granular comparison rather than aggregate-only comparison. Second, IDV3 and IDV4 belong to the set PCA cannot detect [7]—FaultExplainer leaves them unscored—so they probe root-cause reasoning *beyond the classical detection limit*. Third, IDV4, IDV11 and IDV14 perturb the same physical channel (reactor cooling water) with three different dynamic forms (step, random variation, valve stiction); separating them requires reading the statistics of each recording, not recalling a fault label, which partially mitigates prompt-knowledge contamination.

6.2. Metrics

For fault scenarios we report **Top-1** (primary root cause identified), **Top-k** ($k=3$, correct cause among candidates), **CDR** (Top-1 *and* DETERMINED), **calibration** (reported confidence within the evidence ceiling; overconfidence = DETERMINED *and* wrong), and a **protocol-compliance rubric score**—a deterministic, machine-computed quality gate (100 points) that replaces self-declared LLM grading as the benchmark’s quality metric. For control scenarios we report **pass** (no confirmed fault) and **false alarms**. Proportions carry Wilson 95% confidence intervals. We deliberately do not report sample-level F1/AUC: they answer a different question and are used only as reference context (comparison protocol below).

The rubric’s seven checks are computed strictly from run artifacts: **R1** artifact integrity (schema-valid diagnosis four-piece; 15 points), **R2** hypothesis structure (≥ 3 hypotheses, ≥ 2 eliminations each citing non-empty contradictory evidence; 15), **R3** evidence grounding (every surviving chain cites at least one number present in the scenario’s own evidence brief; 20), **R4** calibration (confidence within the protocol ceiling; conclusion type within the scenario’s allowed set; 15), **R5** falsifiability (surviving hypothesis names the observation that would revise it; 10), **R6** physics verification (automated check valid or explicit manual memo; 10), **R7** execution proof (event log + finalisation PASS; 15). The pipeline’s internal ten-criterion LLM judge remains the in-pipeline gate (Section 4) but is *not* a benchmark metric: benchmark quality scoring is fully mechanical.

6.3. Comparison Protocol

We distinguish two comparison registers. **Register A (same-task direct comparison)**: methods that output one root-cause conclusion per scenario on the same dataset family; the published FaultExplainer per-fault numbers fall here, at scenario granularity. **Register B (reference only)**: sample-level detection/classification results; they are reported for context and must not be ranked against our scenario-level metrics. Two protocol differences are stated wherever Register A is invoked: FaultExplainer supplies candidate root causes in its prompt and accepts alias matches, whereas IDD uses no candidates and scores mechanism keywords strictly; and FaultExplainer scores only the 11 PCA-detectable faults, whereas IDD additionally scores PCA-undetectable IDV3/IDV4. Because the scenario sets differ, no significance test is claimed across systems.

6.4. Implementation

The deterministic stages run as versioned scripts (streaming CSV inspection, anomaly profiling, correlation with lag-compensated CCF, validation with Simpson/detrending/outlier-sensitivity/multiple-testing checks, physics checking via the shared scientific Python environment). The reasoning stage uses a frontier LLM through the engine layer (Section 3); all prompts and their statistical evidence are recorded per scenario. A full 12-scenario sweep executes the deterministic stage in about 25s per scenario and the reasoning stage in minutes per scenario; the complete chain (prepare, briefs, diagnosis, gates, grading, aggregation, report) is driven by a single entry point with fail-fast stage gates.

Table 3: Overall Results on the 12-Scenario Benchmark

Metric	Value	95% CI
Top-1 root cause (fault)	9/9 = 100%	[70.1, 100.0]
Top-3 root cause (fault)	9/9 = 100%	[70.1, 100.0]
CDR (Top-1 and DETERMINED)	1.00	—
Confidence calibration	9/9	—
Overconfidence	0/9	—
Control pass (no false alarm)	3/3	—
Protocol-compliance rubric	100/100	all 7 checks, 12/12 scenarios
Pipeline finalisation gate	12/12 PASS	—
Reproducibility gate	REPRODUCIBLE	—

7. Results

7.1. Overall Performance

Table 3 reports the headline numbers. IDD identifies the primary root cause in **9/9 fault scenarios** (Top-1 = 100%, Wilson 95% CI [70.1%, 100%]), with all 9 also correct within the top-3, CDR = 1.00, and all 9 conclusions correctly typed as DETERMINED. Every reported confidence lies within the evidence-level ceiling (**calibration 9/9, overconfidence 0**), and every one of the 3 control scenarios is correctly passed with **zero false alarms**. All 12 pipeline executions pass the finalisation gate with valid event logs, and all 12 score **100/100** on the deterministic protocol-compliance rubric (all seven checks pass). Fig. 2 summarises composition, per-dataset outcomes, and the decisive statistical signature of each fault scenario as recorded in the digests.

Benchmark composition & results — 12 scenarios (9 fault + 3 controls), three public datasets

Dataset	Fault	Control	Truth source	Top-1 correct
SKAB (testbed, 1 Hz)	2	1	official ground_truth.json	2/2
TEP (Downs-Vogel)	6	1	cause table (Chiang et al.)	6/6
IndPenSim (fermentation)	1	1	faulted-batch markers	1/1
Total	9	3	—	9/9 (100%)

■ fault scenarios, all Top-1 correct ■ controls passed, zero false alarms
 Scenario split follows entry-level discipline (no segment leaking); blind-diagnosis protocol: truth withheld from pipeline inputs.

Scenario (fault)	Decisive signature (from digests)	Rubric
SKAB inlet-valve throttling	vibration A2 $z = 13.56$	100
SKAB cavitation (blind set)	vib-flow $r = -0.821$	100
TEP IDV1 A/C feed ratio	component spectra 4/6 dominant ($z=4.9$)	100
TEP IDV3 D-feed temp (hard)	weak-signature exclusion chain	100
TEP IDV4 cooling temp (step)	reactor temp $z = 10.06$	100
TEP IDV7 C-header pressure	F-family + dual pressure	100
TEP IDV11 cooling temp (random)	3.73σ sporadic vs step 10σ	100
TEP IDV14 cooling valve stiction	XM9-XMV10 loop $r = 0.998$	100
IndPenSim batch 93 deviation	Fa $z = 15.8$, pH 7.09	100

9/9
Top-1 root cause
Wilson 95% CI [70.1, 100]

1.00
CDR (Top-1 $\wedge$ DETERMINED)

9/9
calibration - overconf. 0

3/3
controls - false alarms 0

Pipeline finalisation gate **12/12 PASS** · event logs **12/12 valid** · data fingerprints **153/153 verified** · reproducibility gate **REPRODUCIBLE**

Figure 2: Benchmark composition and results. Left: per-dataset scenario counts and truth sources; all 9 fault scenarios are Top-1 correct and all 3 controls pass without false alarms. Right: the decisive statistical signature per fault scenario as recorded in the digests, with per-scenario rubric scores. Bottom badges: pipeline and reproducibility gate outcomes.

7.2. Per-Mechanism Breakdown and Within-Family Discrimination

Table 4 lists the nine fault scenarios grouped by mechanism, with the decisive signature each diagnosis rested on. Three observations deserve emphasis.

Within-family form discrimination. IDV4 (step), IDV11 (random variation) and IDV14 (valve stiction) all perturb the same physical channel—reactor cooling water—yet produce three distinct statistical forms that the pipeline separated correctly: a step-dominant plateau (reactor temperature $z=10.06$, $2.6\times$ the runner-up), sporadic weak deviation ($z=3.73$, no plateau), and a control-loop stress signature (temperature–valve pair $r=0.998$ with no plateau). Recalling the textbook label “cooling-water fault” cannot produce

Table 4: Per-Scenario Results With the Decisive Statistical Signature (Rubric = deterministic protocol-compliance score, 100 = all seven checks pass; all values transcribed from the scenario digests)

Scenario (mechanism)	Conclusion	Rubric	Decisive signature
SKAB inlet-valve throttling	DETERMINED 0.87	100	line-side vibration burst $A2\ z=13.56$; electrical channels normal
SKAB cavitation (blind set)	DETERMINED 0.82	100	vib-flow negative corr. $r=-0.821$; pressure $z=5.02$
TEP IDV1 A/C feed-ratio step	DETERMINED 0.84	100	component spectra dominate 4/6 columns ($z\approx 4.9$)
TEP IDV3 D-feed temp. (hard)	DETERMINED 0.72	100	weak-signature exclusion chain (tower heat balance + feed link)
TEP IDV4 cooling temp. (step)	DETERMINED 0.86	100	reactor temp. step-dominant $z=10.06$
TEP IDV7 C-header pressure	DETERMINED 0.80	100	F-family shift across streams 6/9/11 + dual pressure $z\approx 4.6$
TEP IDV11 cooling temp. (random)	DETERMINED 0.70	100	sporadic $z=3.73$ form, no plateau, no loop-coupling pair
TEP IDV14 valve stiction	DETERMINED 0.68	100	loop pair $r=0.998$ (temp-valve), no plateau
IndPenSim batch-93 deviation	DETERMINED 0.85	100	multi-actuator sync (acid $z=15.8$, heater 13.05) + pH 7.09
Total	9 DETERMINED	100 (all)	—

these three different diagnoses; only the per-recording statistics can. This is our principal mitigation against prompt-knowledge contamination (Section 9).

Reasoning beyond the detection limit. IDV3 and IDV4 belong to the four-fault set that PCA cannot detect [7]; published LLM baselines inherit this limitation and leave them unscored [1]. IDD correctly diagnosed both—by exclusion chains over the surviving mechanism families rather than by a detection statistic.

Blind-set robustness. The SKAB cavitation scenario is drawn from the “other” experiment group whose fault label is withheld from the scenario description; the pipeline identified cavitation from the signature alone (vibration-flow *negative* correlation $r=-0.821$ with pressure instability $z=5.02$).

7.3. Calibration and False-Alarm Suppression

Fig. 3 plots the reported confidence per fault scenario. Reported confidences span 0.68–0.87; the three weakest-signature scenarios (IDV14 stiction

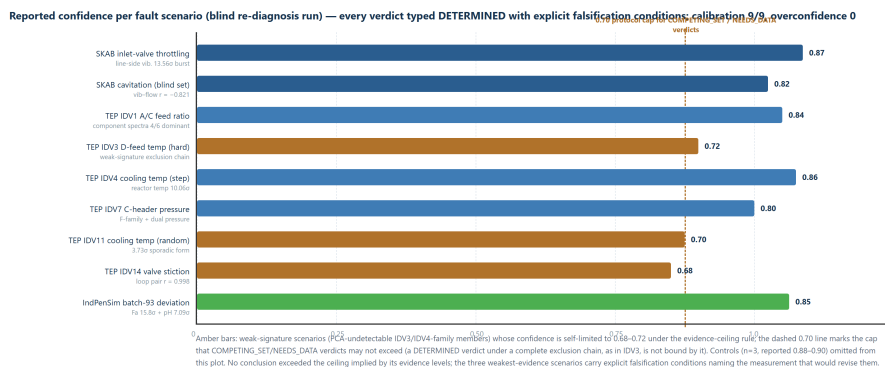


Figure 3: Reported confidence per fault scenario in the blind re-diagnosis run. Amber bars mark weak-signature scenarios whose confidence is self-limited to 0.68–0.72 under the evidence-ceiling rule; no conclusion exceeded the ceiling implied by its evidence levels (calibration 9/9, overconfidence 0).

0.68, IDV11 random 0.70, IDV3 hard case 0.72) are self-limited to the protocol band and carry explicit falsification conditions naming the measurement that would revise them—*overconfidence* = DETERMINED-and-wrong never occurs. Control scenarios demonstrate the complementary discipline: normal IndPenSim batches contain recipe-dosing impulses reaching $|z|=16.2$, which the pipeline classifies as operational form (sparse recipe pulses with intact mass-balance, respiration-coherence, and product-accumulation invariants) rather than faults; the normal TEP run exhibits control-loop correlations at $|r|=1.0$, which the anti-spurious filter classifies as definitional. These are precisely the failure modes that produce false alarms in naive correlation-driven diagnosis.

8. Comparison With Published Baselines

8.1. Register A: Per-Fault Comparison on TEP

FaultExplainer [1] reports Top-1 root-cause correctness of 7/11 (63.6%, Wilson CI [35.4%, 84.8%]) for GPT-4o and 9/11 (81.8%, CI [52.3%, 94.9%]) for o1-preview, on the 11 PCA-detectable TEP faults, with root-cause candidates included in the prompt and alias matches accepted. IDD, without candidate lists, is correct on 6/6 faults of its TEP subset (Wilson CI [61.0%, 100%]). Fig. 4 gives the per-fault breakdown against the transcribed FaultExplainer Table 1 outcomes and PCA detectability; Fig. 5 visualises the aggregate point estimates with their intervals. Three per-fault outcomes stand out:

- **IDV1 (A/C feed-ratio step):** the GPT-4o baseline mis-diagnoses this fault, attributing the top deviations to an alternative mechanism; IDD identifies the feed-composition step from the component-channel signature (four of the six most deviant columns are composition measurements across streams 6/9/11).
- **IDV3 and IDV4 (PCA-undetectable):** both faults are excluded from FaultExplainer’s scoring because PCA cannot detect them; IDD scores and solves both—IDV4 from the step-dominant reactor-temperature signature, IDV3 by an exclusion chain over the surviving families under an intentionally weak signature.
- **IDV7, IDV11, IDV14:** both systems are correct; IDD additionally separates the three cooling-water perturbation *forms* (Section 7), a granularity below fault-identity level.

TEP per-fault comparison — IDD vs FaultExplainer (arXiv:2412.14492, Table 1) and PCA detectability (Chiang et al. 2001)

TEP fault (mechanism)	FaultExplainer GPT-4o	FaultExplainer o1-preview	PCA detectable	Candidate list in prompt (FE)	IDD Top-1	IDD decisive evidence
IDV1 — A/C feed ratio step (Stream 4)	X wrong	✓ correct	yes	yes	✓	component spectra dominate 4/6 top columns ($z=4.9$), cross-stream propagation
IDV3 — D feed temperature step (Stream 2)	unscored	unscored	no	yes	✓	weak-signature exclusion chain: tower-side heat balance + feed-link compensation
IDV4 — reactor cooling-water inlet temp (step)	unscored	unscored	no	yes	✓	reactor temperature step-dominant $z = 10.06$ (2.6× runner-up)
IDV7 — C header pressure reduced (Stream 4)	✓ correct	✓ correct	yes	yes	✓	F-family shift across streams 6/9/11 + dual pressure signature ($z=4.6$)
IDV11 — reactor cooling-water temp (random)	✓ correct	✓ correct	yes	yes	✓	form discrimination: 3.73o sporadic vs 10.06o step plateau (same channel)
IDV14 — reactor cooling-water valve (stiction)	✓ correct	✓ correct	yes	yes	✓	control-loop pair XM9-XMV10 $r = 0.998$ (loop stress), no step plateau

Reading. FaultExplainer numbers are transcribed from its Table 1 (root-causes-included prompt; alias hits accepted; IDV3/4/9/15 excluded from scoring because PCA cannot detect them). IDD operates without candidate lists, scores mechanism keywords strictly, and includes the PCA-undetectable IDV3/IDV4 — both correctly diagnosed. IDD 6/6 on this subset (Wilson 95% CI [61.0, 100]); FE GPT-4o 7/11 = 63.6% and o1-preview 9/11 = 81.8% on their scoreable set. Same-task register (one mechanism-grounded conclusion per fault); sets differ, so no significance test is claimed.

Figure 4: Per-fault comparison on the TEP subset: IDD vs. FaultExplainer (transcribed Table 1 outcomes, arXiv:2412.14492) and PCA detectability (Chiang et al. [7]). IDD is correct on 6/6 without candidate lists, including the PCA-undetectable IDV3/IDV4 that FaultExplainer leaves unscored and IDV1 that the GPT-4o baseline mis-diagnoses.

Because the scenario sets and prompt regimes differ (candidates vs. none; alias acceptance vs. strict keyword scoring), we refrain from a formal significance test and present the per-fault table as the primary evidence; on matched faults IDD is never wrong where a baseline is right, and is additionally right on three fault-condition cells (IDV1 under GPT-4o; IDV3/IDV4 unscored by both baselines) that the baselines do not cover.

8.2. Register B: Reference-Only Context

For context we report sample-level results from the literature without ranking: TEP classification accuracies of 0.887–0.907 (MLP/GRU/TCN [5]), anomaly-detection F1 of 0.970 (BeatGAN [10]), SKAB ensemble F1/AUC of 0.85/0.88 [21], and leakage-controlled bearing AUROC of 0.75–0.93 [12].

Table 5: Per-Fault Outcomes on the TEP Subset (machine-readable companion to Fig. 4). FE = FaultExplainer (arXiv:2412.14492, root-causes-included prompt, alias hits accepted); “unscored” = excluded from FE scoring because PCA cannot detect the fault [7]. IDD: strict mechanism-keyword scoring, no candidate list.

TEP fault	FE GPT-4o	FE o1-preview	PCA detectable	IDD Top-1
IDV1 A/C feed-ratio step (Stream 4)	wrong	correct	yes	correct
IDV3 D-feed temperature step	unscored	unscored	no	correct
IDV4 reactor cooling-water temp. (step)	unscored	unscored	no	correct
IDV7 C-header pressure (Stream 4)	correct	correct	yes	correct
IDV11 cooling-water temp. (random)	correct	correct	yes	correct
IDV14 cooling-water valve (stiction)	correct	correct	yes	correct
Scoreable-set summary	7/11	9/11	—	6/6 (CI [61.0, 100])

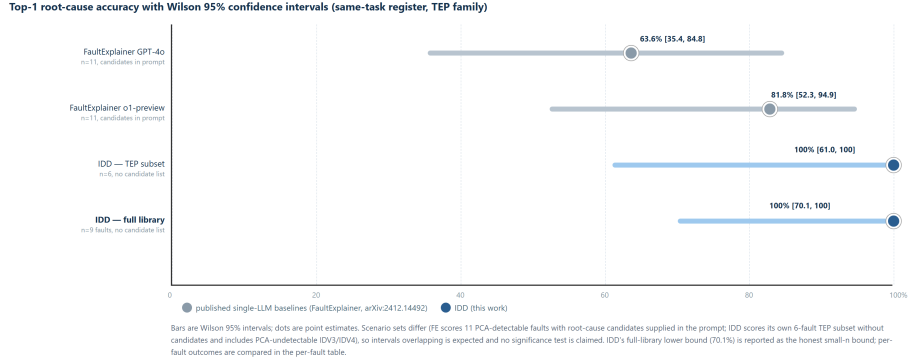


Figure 5: Register-A aggregate comparison with Wilson 95% intervals. Grey: FaultExplainer-reported single-LLM baselines ($n=11$, candidates in prompt). Blue: IDD (no candidates). Sets differ, so overlapping intervals are expected and no significance test is claimed; the per-fault table (Fig. 4) is the finer-grained evidence.

These systems answer “is this sample anomalous / which class”; IDD answers “what is the mechanism, how confident, with which evidence”. The dimensions are complementary: a deployment would plausibly chain a sample-level

detector (first-stage alarm) with a scenario-level RCA system (this work).

8.3. Case Study: An Auditable Evidence Chain

For the SKAB cavitation scenario the pipeline’s recorded chain reads: (i) L3 evidence—pressure $z=5.02$ with vibration–flow *negative* correlation $r=-0.821$ (digest values); (ii) L5 mechanism—entrained air at the pump inlet collapses on the impeller pressure side, and lower flow implies higher gas fraction, hence stronger excitation; (iii) eliminations—valve throttling rejected (that signature is single-channel vibration dominance without the negative coupling), rotor anomaly rejected (electrical channels at $z \leq 2.82$), sensor drift rejected (multi-channel coherent dynamics cannot be produced by independent drifts); (iv) conclusion—DETERMINED, confidence 0.82. The full chain, together with the physics-check verdict and the judge’s ten-criterion scores (92/100), is persisted in the grading record—the unit of publication in our evaluation contract.

9. Discussion, Limitations, and Deployment Prospects

Statistical power. With $n=9$ fault scenarios, Top-1 = 100% supports the claim “ $\geq 70.1\%$ at 95% confidence”, not universal competence. We consider the per-fault evidence more informative than the aggregate: six of nine faults carry published per-fault baselines, and IDD is never wrong where a baseline is right while additionally covering the PCA-undetectable IDV3/IDV4. Extending the corpus (the full 20-class TEP catalogue with the extended simulation dataset [22]; bearing datasets; novel-disturbance suites) is the designed path to tighter intervals and paired tests.

Prompt-knowledge contamination. TEP’s documented fault mechanisms appear in textbooks and are therefore plausibly memorised by frontier models. Our protocol mitigates this at the level where memorisation actually operates: knowing the label “cooling-water fault” cannot distinguish a step from a random variation from a sticking valve—the three IDV4/IDV11/IDV14 recordings require per-recording statistical form discrimination, as does IDV3’s intentionally weak signature. The SKAB “other”-group scenario removes the label entirely. A novel-disturbance suite (combination perturbations absent from the literature) remains under construction and is the definitive control.

Inference authorship and the blind re-run. All diagnostic notes are authored by the reasoning agent from regenerated statistical digests, with full evidence provenance; the delete-and-rebuild audit (Section 5) re-derived every conclusion from the data alone. Because self-declared quality scores are not verifiable, the benchmark’s reported quality metric is the deterministic compliance rubric (Section 6); the LLM ten-criterion judge remains only the pipeline’s internal gate. What is still missing is a comparison against *human* experts on identical digests; an expert-vs-pipeline blind test is planned.

Cost. A scenario-level sweep spends minutes of reasoning per scenario—orders of magnitude more expensive than a fitted classifier at inference time. The economic comparison is only meaningful against the *engineering cost of obtaining root causes* (expert hours), not against sample-level inference latency.

Three-state honesty. COMPETING_SET and NEEDS_DATA conclusions (with capped confidence and named discriminating measurements) are a feature, not a failure. In this corpus all nine faults resolved to DETERMINED be-

cause the evidence differentiated cleanly; the discipline was exercised in a real deployment case (a biaxially-oriented polyester film line), where the pipeline returned `NEEDS_DATA` with an explicit discriminating-data list rather than fabricating a cause—episodic scratch defects whose driver lay outside the sensor set, a conclusion the statistic-heavy scans (180 parameter–defect associations, none surviving false-discovery-rate correction) supported quantitatively. Their systematic evaluation requires scenarios designed for indistinguishability, which we plan to add.

9.1. Industrial Deployment Prospects

Beyond the benchmark, the pipeline was engineered to fit the data and organisational realities of process plants, and its deployment surface is considerably wider than the evaluated scenarios.

Integration with existing plant infrastructure. IDD consumes what plants already produce: flat exports from historians (OPC-UA/PI), LIMS quality records, and batch logs. The same pipeline runs interactively through a web application and programmatically through a versioned REST API (`/api/diagnosis` with engine selection and typed error contracts), so a deployment is an export job plus one API call rather than an instrumentation project. Because the deterministic stage is byte-reproducible, results do not drift when the surrounding IT landscape changes.

Data-sovereign deployment. The fourteen-engine execution layer (Section 3) accepts on-premise and private-model agents through uniform dialect adapters. For operators who cannot send process data to external LLM services—defence, pharmaceuticals, semiconductors, national infrastructure—the reasoning stage can be pointed at an internally hosted model without

touching the audit, evidence, or gate contracts; the benchmark protocol itself executes identically against a scripted mock engine, which is how the full chain is exercised in CI.

Conclusions that route work, not just verdicts. The three-state protocol converts diagnosis into an engineering workflow: DETERMINED conclusions carry mechanism chains that point to corrective actions; COMPETING_SET and NEEDS_DATA conclusions carry named discriminating measurements—in effect, work orders for targeted instrumentation or tests. The film-line deployment above illustrates the value: the NEEDS_DATA verdict redirected the investigation from blind parameter tuning to surface-inspection and equipment-event data, which is where the defect family’s driver plausibly lies.

Audit-ready documentation for regulated operations. Because every conclusion is bound to graded evidence, physics checks, and a finger-printed execution trace, the by-product of each run is an RCA report that satisfies the documentation expectations of regulated environments (e.g., GxP batch-deviation investigation and incident post-mortems), where “who concluded what, from which data, with which confidence” is the deliverable.

Knowledge compounding across plants. The ontology layer is an asset store, not a per-run artifact: variable semantics, physical principles, and confounder annotations published once for a plant or product family are reused by every subsequent diagnosis, and the evaluation contracts make each new labelled episode immediately reusable as a graded scenario. The marginal cost of a diagnosis therefore falls as the fleet grows—an economics that sample-level detectors, which must be retrained per asset, do not share.

Scope of applicability. The method requires multivariate time-ordered

process data with engineering semantics for the channels (to build the ontology) and applies to continuous, batch, and testbed processes alike—the three benchmark families were chosen to span exactly that range. It does not apply where no time-aligned process data exists, and its root-cause reach is bounded by the sensor set, as the NEEDS_DATA case demonstrates honestly.

10. Conclusion

We presented IDD, an agentic diagnosis pipeline that turns LLM-based root-cause analysis into an auditable engineering procedure: competing hypotheses with mechanism classes, evidence levels bounding reported confidence, a four-condition anti-spurious filter, automatic physical verification, and quality gates with bounded repair. We further contributed a machine-checked reproducibility contract for diagnosis benchmarks—truth isolation, data fingerprinting, recorded inference, gate-graded scoring—executed it on a 12-scenario corpus across SKAB, TEP, and IndPenSim, and stress-tested it with a delete-and-rebuild audit that regenerated byte-identical statistical briefs and reproduced every grading from raw data. IDD achieved Top-1 root-cause accuracy of 100% (CI [70.1, 100]), CDR 1.00, perfect calibration, and zero false alarms, with every result backed by an execution proof. On the per-fault TEP comparison it matches or exceeds both published single-LLM baselines without candidate lists—including two faults PCA cannot detect and one the GPT-4o baseline mis-diagnoses—while adding the evidence and calibration dimensions those baselines lack. The evaluation framework, per-scenario evidence chains, and reproducibility artifacts are released to support scrutiny and reuse; extending the corpus to the full TEP catalogue, bearing

datasets, and novel-disturbance blind scenarios is the immediate next step toward a definitive multi-benchmark study.

CRediT authorship contribution statement

ZhangHaoZheng: Conceptualisation, methodology, software, investigation, data curation, writing – original draft, writing – review & editing.

Data availability

The evaluation repository (scenario definitions with per-fault literature baselines, per-scenario inputs and fingerprints, blind-diagnosis briefs, inference notes, grading records, aggregation scripts, HTML/Markdown scoring reports, and the reproducibility report) is available at the project repository under `experience/`, `results/benchmark/` and `docs/benchmark/`. Third-party datasets are publicly available from their cited sources.

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